A03 / INDIVIDUAL CONTRIBUTION

Notebook Author, Layout & Submission Lead

Team Member: Michael Rivas | Team: Team 3 | Course: ITAI 2376 - Deep Learning | Assignment: A03 Neural Network Zoo

Role in one sentence: I wrote the integrated notebook, compiled the team's findings into it, designed its layout and zoo tour, and handled submission and direct communication with the instructor.

1. Introduction & Objectives

My job was to turn separate team findings into one submission a grader can actually follow and run. That meant writing a single notebook that, in order, introduces neural networks, shows the zoo/safari chart, lets a group choose an animal, presents the combined tour, and ends with reflection - and making sure the visual layout, the code, and the written explanations all said the same thing.

The objective was not to add another isolated report. It was to make the team's work findable, in one place, with each architecture, its mechanism, and its typical use clearly labelled, and with a working program behind the pictures.

2. What I Wrote and Built

2.1 The integrated notebook

I authored A03_Neural_Network_Safari_Deck_Notebook.ipynb as the team's submission notebook. I wrote and assembled, in order:

- Section 1 - Introduction: neurons, weights, bias, activation functions, layers, and the forward pass, in plain language.
- Section 2 - The Safari chart: seven animal-architecture pairings - Ostrich / FNN-MLP, Leopard / CNN, Hyena / RNN, African Grey Parrot / LSTM, Octopus / Autoencoder, Spider / GNN, and Honeybee / Transformer - each with mechanism, analogy, and typical applications.
- Section 3 - Interactive group activity: the choose-an-animal exhibit, with Spider / GNN and Honeybee / Transformer as the interactive options, and a rendered exhibit card.
- Section 4 - Presentation & Zoo Tour: the combined gallery and animated carousel that bring all the exhibits together for the class tour.
- Section 5 - Reflection: similarities across architectures, differences, the odd one out, and where each architecture fits - or does not fit - a task.
- Section 6 - Computational demonstration: the Leopard 1D CNN and Parrot LSTM on the same synthetic burst-detection task, with training loss and test accuracy reported from the saved notebook output.

2.2 Layout and visual design

I designed the layout so the safari reads as one expedition, not as seven disconnected posters: a single expedition map, consistent animal cards with route, habitat, mechanism, applications, and data flow, then a gallery in the same style. I kept the animal work as an analogy, with an explicit note in the notebook that animal behaviour does not literally model a neural network - it is a teaching aid for data flow, memory, and feature extraction.

I also compiled the team's findings into that layout, so a reader does not have to hunt across separate files to see how the data work, the architecture work, and the exhibits work relate to each other.

2.3 Compilation of team findings

The notebook brings together work from the whole team, with credit kept separate in each member's own contribution PDF:

- Cherluk - Data Engineering & Preprocessing: data pipeline, tensor formatting, and validation before training.
- Olu - Neural Architecture & Computational Demonstration: the Leopard CNN and Parrot LSTM implementations and their recorded results (CNN 98.00%, LSTM 48.50% on the 200-example test split).
- Taki - Interactive Exhibits, Gallery & Evaluation: how students explore exhibits, how the gallery is assembled, and how recorded results should - and should not - be interpreted.

My contribution was writing, integrating, and laying out the notebook that contains those findings, not claiming their specialised work as if I had done it alone. Where a claim in the notebook depends on a teammate's work, that work is named in this section and in their own PDFs.

3. Submission and Instructor Communication

I was the team's direct contact with the instructor throughout this assignment, and the submitter named in the Canvas files (A03_Team_3_Rivas_ITAI2376.mp4 and A03_Rivas.pdf). I handled the Canvas submission and the follow-up when the instructor asked, on Sep 12, Sep 23, and Sep 27, for a specific git repository link posted in Canvas, with PDF files only in the repository, rather than documents sent by email.

This PDF is written for that repository: it states, in one findable document, what my contribution was, so the instructor does not have to search the notebook, the video, and separate team files to locate it. The video already submitted in Canvas documents the project presentation; this PDF documents my individual part in building it.

4. Challenges & Solutions

4.1 Keeping pictures, code, and text consistent

A visual project can easily drift: a card can say one thing, the code can do another, and the gallery can show a third version. I kept one set of animal names, architectures, and mechanisms across the map, the cards, and the tour, and I checked the notebook's saved outputs against the text before submission.

4.2 Making the result honest, not just impressive

The computational demonstration does not flatter every model. The LSTM's 48.50% result is left in, with the interpretation that a short local burst task favours a convolutional model with pooling, while a recurrent model reading to the last step has to carry an early signal through later noise. That distinction - analogy versus measured result - is part of the reflection, not an error to hide.

4.3 Delivery, not just content

The hardest part of this submission was not writing the notebook, it was making it findable in the format the instructor asked for. A good notebook that cannot be located in a repository, or a contribution that is only visible inside a video, does not meet the submission instruction. This

repository package, with PDF files only and this contribution clearly named, is the fix for that specific problem.

5. Reflection & Key Insights

Writing the whole notebook taught me that integration is itself a technical skill. Choosing an animal is easy; showing, in the same document, how its mechanism maps to a layer, a state, a gate, or a bottleneck - and then showing a program that actually runs - is where understanding is tested.

Key insight: A zoo is useful only if the tour has a map. Without a consistent layout, shared labels, and a clear order from introduction to reflection, even correct individual pieces look disconnected. With them, a reader can compare architectures, spot mismatches, and see why no single animal fits every task.

I also learned to separate what is proven from what is illustrated. The safari pictures illustrate ideas. The CNN and LSTM numbers are the only trained-model measurements in the demonstration, and they describe one synthetic experiment, not a general ranking of CNNs versus LSTMs.

6. Supporting Evidence & Attribution

- Notebook: A03_Neural_Network_Safari_Deck_Notebook.ipynb - Sections 1-6 as listed above, including the saved CNN 98.00% / LSTM 48.50% output and the zoo tour gallery.
- Canvas submission files: A03_Team_3_Rivas_ITAI2376.mp4 (project video) and A03_Rivas.pdf (this contribution, as submitted).
- Team contribution PDFs: Cherluk_A03_Finalized.pdf, Olu_A03_Finalized.pdf, Taki_A03_Finalized.pdf - each member's own account of their specialised work.
- Instructor correspondence: Sep 12, Sep 23, and Sep 27 requests for a specific git repository link in Canvas, PDF files only, not by email.

Scope note: This report describes the notebook and submission as assembled by me. It does not claim that every teammate's specialised claim has been re-run by me, and it does not claim a repository link existed before this package was prepared - that link is the remaining delivery step, to be posted in Canvas once the repository is created.